

The artifact I chose is my CS 360 Android weight tracking application, which I originally created in 2024. The app allows users to create accounts, log in, track their weight, and use a guest account.

I selected this artifact because it shows my skills in Java, Android development, software design, and database management. I improved the application by adding the `UserSession` and `Weight Record` classes, properly connecting users to their unique IDs, separating guest and registered user data, and improving how weight records are stored and deleted.

I met the course outcomes I planned to address by applying software engineering and design principles to improve the existing application. I do not currently have any changes to my outcome coverage plans, and I will work on more outcomes as I continue with enhancements 2 and 3.

I learned more about organizing an application into separate classes and properly managing user data and sessions. One challenge was improving the existing login and database functionality without breaking the applications original features, which was where a lot of my testing and debugging went into. I also had to fix the `activity_data_display.xml` file so it looks cleaner and easier to use than before.

Bugs that were fixed with this enhancement:

- There was a missing intent, and every user, registered or guest, read and wrote `user_id-1`
- `db.close()` happened after every helper method.
- `GetWeightData()` returned a live cursor, then `loadDataIntoGrid()` iterated it on a connection other methods already closed

-The `getWeightDataID()` threw `ArrayIndexOutOfBoundsException` on any weight without a “(“, and returned the wrong ID whenever two entries matched.